

Lecture 7: Normal Distribution, Joint PDFs

Jamie Haddock

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- 1 The Normal distribution::: **{.callout-note icon=false}## Definition: Normal / Gaussian** $X \sim N(\mu, \sigma^2)$ **has PDF**

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}.$$

Standardizing $Z = (X - \mu)/\sigma$ **gives** $Z \sim N(0, 1)$.:::—**## Joint PDFs::: {****.callout-note icon=false}## Definition: Jointly continuous random variables** X, Y **are jointly continuous with PDF** $f_{X,Y} \geq 0$ **if**

$$P((X, Y) \in R) = \iint_R f_{X,Y}(x, y) dx dy,$$

and $f_X(x) = \int f_{X,Y}(x, y) dy$.:::—**## Bivariate Normal: theory vs. simulation** 1

—title: “Lecture 7: Normal Distribution, Joint PDFs”author: “Jamie Haddock”format: revealjs: output-file: Lecture7_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
mean = [0, 0] cov = [[1, 0.6], [0.6, 1]] samples = rng.multivariate_normal(mean, cov, size=5000) plt.scatter
```

Next time: Bayes' rule for random variables, and a preview of Bayesian estimation.## Reading & Homework
*Reading: §3.3-3.5 (Bertsekas & Tsitsiklis, Introduction to Probability).***Due today:** Homework 3 (Gradescope).